

Program : Diploma in Printing Technology	
Course Code : 3106	Course Title: Material Testing Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To make students acquainted with all the physical and chemical tests of materials used in different printing process.

Course Prerequisites:

Topic	Course code	Course name	Semester
Caliper Test		Applied Physics Lab-I	I
pH test, Conductivity Test		Applied Chemistry Lab-I	I

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate Paper Testing methods.	11	Applying
CO2	Demonstrate Fountain solution Testing methods.	10	Applying
CO3	Demonstrate Ink Testing Methods.	10	Applying
CO4	Demonstrate Miscellaneous materials Testing.	10	Applying
	Lab Exam	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate Paper Testing methods.		
M 1.01	GSM test	4	Applying
M 1.02	Grain Detection test	3	Applying
M 1.03	Moisture content of paper Moisture penetration of paper	4	Applying
CO2	Demonstrate Fountain solution Testing methods.		
M 1.01	Introduction to pH meter and conductivity meter	3	Applying
M 1.02	pH	3	Applying
M 1.03	Conductivity	4	Applying
	Lab Exam I	2	
CO 3	Demonstrate Ink Testing Methods.		
M 1.01	Drying time	4	Applying
M 1.02	Flow property	4	Applying
M 1.03	Length	2	Applying
CO 4	Demonstrate Miscellaneous materials Testing.		
M 1.01	Paper board Testing (GSM, Thickness)	5	Applying
M 1.02	Blanket Hardness Testing	5	Applying
	Lab Exam II	2	

Text / Reference:

T/R	Book Title/Author
T1	Tulika Das, Chemistry in Printing, 2nd Edition , Barnana Prakashani, 2011.
R1	P. Kipphan, Handbook of Print Media , Springer, 2002.
R2	Bishwanath Chakravarty, A Hand Book for Printing and Packaging , Galgotia Publications Pvt. Ltd, 1997.
R3	GATF Staff Solving Offset Ink Problems GATF, 1998
R4	N.R.Elred & T. Scarlet, What the Printer should know about Ink , GATF, 1995
R5	N.R.Elred & T. Scarlet, Chemistry for the Graphic Arts , GATF, 1992.

Online Resources:

Sl.No	Website Link
1	https://www.trl.com/electrical-properties-testing/
2	https://www.sciencedirect.com/topics/engineering/conductivity-measurement
3	https://www.texcareinstruments.com/gsm-testing-machine.html
4	https://www.defelsko.com/resources/offset-printing-blanket-thickness-measurement